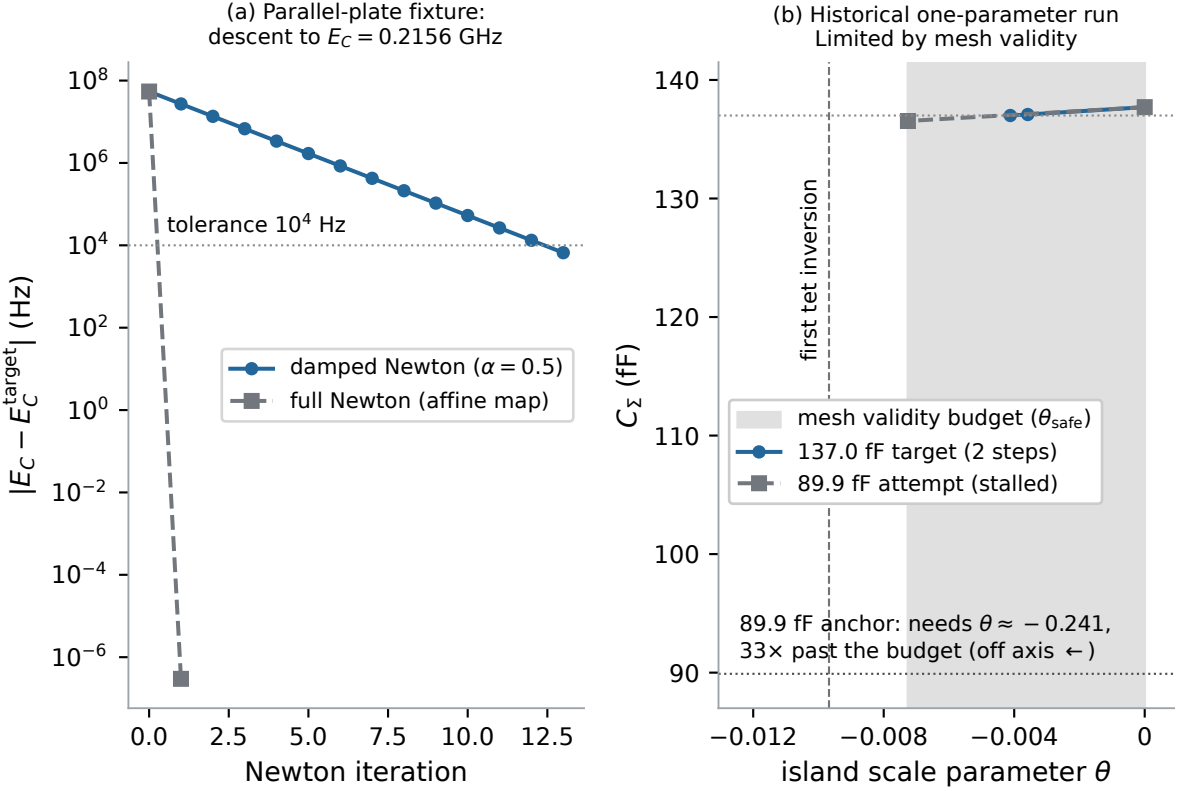
(a) Parallel-plate fixture:
descent to $E_C = 0.2156$ GHz
$|E_C - E_C^{\text{target}}|$ (Hz)
10^8
10^6
10^4
10^2
10^0
10^{-2}
10^{-4}
10^{-6}
tolerance 10^4 Hz
damped Newton ($\alpha = 0.5$)
full Newton (affine map)
0.0
2.5
5.0
7.5
10.0
12.5
Newton iteration
(b) Historical one-parameter run
Limited by mesh validity
C_Σ (fF)
140
130
120
110
100
90
first tet inversion
mesh validity budget (θ_{safe})
137.0 fF target (2 steps)
89.9 fF attempt (stalled)
89.9 fF anchor: needs $\theta \approx -0.241$,
33× past the budget (off axis ←)
−0.012
−0.008
−0.004
0
island scale parameter θ